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Your Roll No.....

Sr. No. of Question Paper : 1427

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Unique Paper Code : 32351303

Name of the Paper : BMATH 307 – Multivariate
Calculus

Name of the Course : B.Sc. (H) Mathematics

Semester : III

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

Deshbandhu College Library
Kalkaji, New Delhi-19

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. All sections are compulsory
3. Attempt any Five questions from each section. All questions carry equal marks

SECTION I

1. Let $f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2}$ if $(x, y) \neq (0, 0)$
 $= 0$ otherwise

Show that $f(0, y) = -y$ and $f(x, 0) = x$ for all x and y .

P.T.O.

2. Use incremental approximation to estimate the function $f(x, y) = \sin(xy)$ at the point

$$\left(\sqrt{\frac{\pi}{2}} + .01, \sqrt{\frac{\pi}{2}} - .01 \right)$$

3. If $z = xy + f(x^2 + y^2)$, show that $y \frac{\partial z}{\partial x} - x \frac{\partial z}{\partial y} = y^2 - x^2$.

4. Assume that maximum directional derivative of f at $P_0(1, 2)$ is equal to 50 and is attained in the direction towards $Q(3, -4)$. Find ∇f at $P_0(1, 2)$.

5. Find the absolute extrema of $f(x, y) = 2x^2 - y^2$ on the disk $x^2 + y^2 \leq 1$.

6. Use Lagrange multiplier to find the distance from $(0, 0, 0)$ to plane $Ax + By + Cz = D$ where at least one of A, B, C is nonzero.

SECTION II

1. Compute the integral $\int_0^1 \int_x^{2x} e^{y-x} dy dx$ with the order of integration reversed.

2. Use Polar double integral to show that a sphere of radius a has volume $\frac{4}{3}\pi a^3$.

3. Compute the area of region D bounded above by line $y = x$, and below by circle $x^2 + y^2 - 2y = 0$.
4. Find the volume of the solid bounded above by paraboloid $z = 6 - x^2 - y^2$ and below by $z = 2x^2 + y^2$.
5. Evaluate $\iiint_D \frac{dx \, dy \, dz}{\sqrt{x^2 + y^2 + z^2}}$, where D is the solid sphere $x^2 + y^2 + z^2 \leq 3$.
6. Use a suitable change of variables to find the area of region R bounded by the hyperbolas $xy=1$ and $xy=4$ and the lines $y=x$ and $y=4x$.

SECTION III

1. Find the mass of a wire in the shape of curve C: $x = 3 \sin t$, $y = 3 \cos t$, $z = 2t$ for $0 \leq t \leq \pi$ and density at point (x, y, z) on the curve is $\delta(x, y, z) = x$.
2. Find the work done by force

$$\vec{F}(x, y, z) = (y^2 - z^2)\hat{i} + (2yz)\hat{j} - (x^2)\hat{k}$$

on an object moving along the curve C given by $x(t) = t$, $y(t) = t^2$, $z(t) = t^3$, $0 \leq t \leq 1$.

3. Use Green's theorem to find the work done by the force field

$$\vec{F}(x, y) = (3y - 4x)\hat{i} + (4x - y)\hat{j}$$

when an object moves once counterclockwise around the ellipse $4x^2 + y^2 = 4$.

4. Use Stoke's theorem to evaluate the surface integral

$$\iint_S (\text{curl } \vec{F} \cdot \vec{N}) dS$$

where $\vec{F} = x\hat{i} + y^2\hat{j} + ze^{xy}\hat{k}$ and S is that part of surface $z = 1 - x^2 - 2y^2$ with $z \geq 0$.

5. Use divergence theorem to evaluate the integral

$$\iiint_S \vec{F} \cdot \vec{N} dS \quad \text{where} \quad \vec{F}(x, y, z) = (\cos yz)\hat{i} + e^{xz}\hat{j} + 3z^2\hat{k},$$

where S is hemisphere surface $z = \sqrt{4 - x^2 - y^2}$ together with the disk $x^2 + y^2 \leq 4$, in x - y plane.

6. Evaluate the line integral $\int_C \vec{F} \cdot d\vec{R}$

Where $\vec{F}(x, y) = [(2x - x^2y)e^{-xy} + \tan^{-1} y]\hat{i} +$

$\left[\frac{x}{y^2 + 1} - x^3 e^{-xy} \right]\hat{j}$ and C is the ellipse $9x^2 + 4y^2 = 36$.